

Radians

Angles, arcs, sectors and segments

YOUR AIM TODAY

Use radians confidently in exact geometry problems.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

The ideas you need

Radians measure an angle using arc length divided by radius.

- π radians = 180 degrees.
- Degrees to radians: multiply by $\pi/180$.
- Arc length $s = r \theta$.
- Sector area = $(1/2)r^2 \theta$.
- Segment area = sector area - triangle area = $(1/2)r^2(\theta - \sin \theta)$.

MEMORY RULE

The arc and sector formulas require θ in radians.

Convert

WORKED STEPS

60 degrees = $60(\pi/180) = \pi/3$.

Sector

WORKED STEPS

$r=6$, $\theta=1.2$: arc = 7.2; area = $0.5(36)(1.2) = 21.6$.

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Convert 135 degrees to radians.	
2. Convert $5\pi/6$ radians to degrees.	
3. Find arc length when $r=8$, $\theta=0.75$.	
4. Find sector area when $r=5$, $\theta=1.4$.	
5. Find the perimeter of that sector.	
6. Find segment area for $r=4$, $\theta=\pi/3$.	

STOP

Do not continue until all six questions have an attempt.

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	$3\pi/4$.
2	150 degrees.
3	$s=8(0.75)=6$.
4	$0.5(25)(1.4)=17.5$.
5	$2r+s=10+7=17$.
6	$0.5(16)(\pi/3-\sin(\pi/3))=8\pi/3-4\sqrt{3}$.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.